

Proxy Features and Fairness

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Subgroup Performance

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Feature Engineering

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Find a proxy risk for Proxy features and fairness. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Proxy features and fairness. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request evidence.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

6. Solve the given case

Background colour tracks collection location, associated with plant types.

Explain proxy risk and test.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A plant classifier uses camera_model, and one species was photographed mostly on a single camera. What shortcut might it learn and how could you investigate?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 32 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Document Plant Classifier proxy risks, paired-test evidence, mitigation, and remaining fairness limitation.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Read an association

What do the device counts support, and what do they not support?

My reasoning and evidence

13. W2 Find a proxy risk

Why might removing building but keeping device code leave related information?

My reasoning and evidence

14. W3 Plan an ablation

Describe a controlled with/without-code comparison.

My reasoning and evidence

15. W4 Apply independently

A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request evidence.

My reasoning and evidence

16. W5 Limit the result

Does unchanged accuracy after removing a proxy prove fairness?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Proxy Features and Fairness

Removing an obvious setting field may leave another field that carries the same information.

Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk. The team drops location.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Proxy Features and Fairness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Remove direct field

The location column is absent from the new input table.



2. Check retained proxy

camera_code still identifies the setting perfectly in this supplied sample.



3. Audit effect

Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

CHANGE ONE THING

Camera codes are assigned independently of setting in a new balanced dataset.

Explain the changed result using the matching lesson.

What evidence contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Removing an obvious setting field may leave another field that carries the same information.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

1. Remove direct field
2. Audit effect
3. Check retained proxy

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Audit effect Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence. Read the picture in order. Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Camera codes are assigned independently of setting in a new balanced dataset. Which conclusion is supported by the supplied facts?

1. camera_code no longer perfectly identifies setting in that dataset.
2. We can decide the result without examining the changed input or the procedure.

3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: camera_code no longer perfectly identifies setting in that dataset. The observed mapping changed; other retained fields may still contain setting information.

Repair a missing building block

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at prediction time. Short practice: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean 70%? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
